CMA Study Notes

Simple Edition: every topic explained in easy words, with an example

Cost & Management Accounting (EPGP, Section C): Days 1 to 4 + Problem Sets 1, 2, 3 + Excel steps

How to read these notes

Yellow box = the idea in simple words. **Purple box** = a worked everyday example. **Blue box** = the professor's method, step by step. **Green box** = how to do it in Excel. **Red box** = exam tips and traps. Read yellow, then purple, then the tables.

Sources covered

Day 1 to Day 4 lecture decks (including the professor's handwritten ink on the slides); Problem Set 1 (with solutions), Problem Set 2, Problem Set 3 (with solutions in the Excel workbook); PlayCraft question with solution.

Corrections to source typos (flagged where they appear)

Set 1 P1 cost of production = 706,625 (file prints 706,620). Set 3 P2 Stores split = 600 and 400 (file prints 6000 and 4000). PlayCraft direct material consumed = 270,000 (file prints 200,000). Set 3 P4 has no printed solution; it was worked from the question and checked (total 26,840). Set 2 Q2 reads the Rs 1.4 lakh DM as DM **used**.

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PART A: Day 1: Introduction to Management Accounting

Big picture in one line

Think of a business as a car and accounting as the dashboard. The dashboard shows speed, fuel and engine heat so the driver can decide what to do next. Management accounting is the dashboard built for the **driver (the manager)**.

Course note from the deck: only 2 quizzes will be held; every class and every assignment carries marks.

1. What an accounting system does

An accounting system takes everyday business events (a sale, a purchase, a salary payment) and turns them into useful information. It does four things to the raw data: **collect, sort (categorise), summarise, analyse**. The result goes into (a) **financial statements** for outsiders and (b) **internal performance reports** for managers.

Example

Chai stall. Every day you note: milk Rs 400, sugar Rs 100, cups sold 300 at Rs 10. That is raw data. At month-end you sort it (milk, sugar, rent), add it up (total cost Rs 21,000, total sales Rs 30,000) and study it (milk cost went up 10%). Now it can help you decide whether to raise the price.

2. Accounting is the "language of business"; different managers need different views

Everyone in a company speaks through numbers: marketing, HR, supply chain, finance all use accounting data (the deck links it to statistics, algebra Y = mX + b, logic, finance, tax, information systems, economics, supply chain, governance and HR). The **same sales data** is read differently by different managers.

Manager	What they want from sales data	Why
Sales manager	Sales by region or by salesperson	To reward and guide the sales team
Distribution manager	Orders by location and due date	To plan delivery trucks
Manufacturing manager	Quantities ordered	To schedule production

Example

An order of 1,000 shirts for Pune due on 5th. Sales manager: "Good, Pune is growing." Distribution manager: "I need a truck to Pune by 4th." Manufacturing manager: "I must make 1,000 shirts by 3rd." One number, three uses.

3. What makes management accounting different

- **Forward-looking:** it is about what *will* happen and what we *should* do.
- **Decision-making focus** (the most important word the professor stresses).
- **Not bound by accounting standards** (Ind AS / GAAP): managers can design any report that helps them.
- **Cost-benefit test:** collect a piece of information only if its benefit is more than the cost of getting it.

Example

Before buying a Rs 20,000 market-survey report, ask: will it help us earn more than Rs 20,000? If not, do not buy it.

4. Management accounting vs Financial accounting

In simple words

Financial accounting = the *report card* shown to outsiders (owners, banks, government): rule-based, about the past, covers the whole company. **Management accounting** = the *game plan* used by insiders: flexible, about the future, detailed by product or department.

Basis	Financial accounting	Management accounting
Period	After a fixed period (quarter / year)	Whenever management needs it, even daily
Objective	Report to owners, creditors, government	Help internal management
Time	Historical (what happened)	Past, present **and future**
Unit	Money only	Money or any unit (kg, hours, units)
Data & sources	Actual past data, internal sources	Future plans and projections + past data; internal and external sources
Coverage	Business as a whole (aggregate)	Limited coverage but in detail (a product, a division)
Standards	Set by outside bodies; objective; money consequences	Set by management itself; subjective; event consequences
Legal need	Compulsory (statutory)	Voluntary (optional)
Precision	"Pie to pie" accuracy (every paisa)	May be an estimate or guess
Purpose	Overview of the whole business	Analytical details to support specific decisions
Unit of account	Whole business recognised	Results of units, divisions
Publication & audit	Essential	Not required
Principles	Ind AS, GAAP, IFRS	No fixed principles; tailored to need

Example

Same company, two questions. The bank asks "What was your profit in 2025?": financial accounting answers with the audited annual report. The owner asks "If I launch a second brand next year, will it make money?": management accounting answers with estimates and scenarios. No law requires the second report, but the owner needs it.

5. Cost accounting vs Management accounting

Basis	Cost accounting	Management accounting
Object	Find the cost of a product	Give information to management
Deals with	Cost data	Cost **and revenue** data
Scope	Limited	Much wider
Nature	Past and present facts	Projection of future plans
Procedures	Fixed procedures exist	No fixed procedures
Data type	Quantitative	Quantitative **and qualitative**
Users	Internal and external	Internal (management)
Value to decisions	Low	High

Example

Cost accounting says "one pizza costs Rs 80". Management accounting uses that Rs 80 plus expected demand, competitor prices and customer taste to decide "sell at Rs 199 or Rs 249? open a new outlet?"

6. Strategy and management accounting

In simple words

Strategy = how a company uses what it is good at to win in the market. There are two broad ways: **cost leadership** (good product, lowest price) or **product differentiation** (special product, higher price). **Strategic cost management** means managing costs with this strategy in mind. Management accountants supply the numbers that show where the advantage comes from: (1) are our costs lower than competitors', or (2) is the premium price we charge bigger than the extra cost of making the product special?

Example

Cost leadership: a discount supermarket keeps costs tiny (basic stores, bulk buying) and sells cheap. **Differentiation:** a premium watch brand spends heavily on design and brand, and charges 5x. The accountant checks: "Does the extra Rs 3,000 price cover the extra Rs 1,200 design cost?" Yes, so differentiation pays.

Answer: QFT 1

Question for thought 1: can a firm follow a strategy "in the middle"? It is risky. A firm that is neither cheapest nor most special gives customers no reason to choose it ("stuck in the middle"). It can work only if the firm has real scale or technology that makes it fairly cheap *and* fairly special. Most winners pick one main position.

Example

A restaurant that is not cheap like a roadside dhaba and not special like a fine-dining place often loses both groups of customers.

7. Managerial accounting as a process

Five verbs: **Identify, Measure, Analyse, Interpret, Communicate** information, always in pursuit of the organisation's goals. It is part of the management process and supplies information for **planning, control and decisions**.

Example

Milk price rises. Identify: milk cost is up. Measure: Rs 50 to Rs 56 per litre. Analyse: cost of one cup rises by Rs 0.60. Interpret: our margin shrinks from 40% to 34%. Communicate: tell the partner and propose raising the cup price by Rs 1.

8. How management accounting adds value (its objectives)

1. **Provide information** for decisions and planning.
2. **Help managers direct and control** activities.
3. **Motivate** managers and staff towards company goals.
4. **Measure performance** of units, activities, managers and employees.
5. **Assess the competitive position** of the organisation.

Example

A branch manager gets a monthly scorecard: sales, wastage, customer complaints. It tells her what to fix (control), gives a bonus target (motivate), and lets the head office compare branches (measure performance).

9. Value, and the three customer value propositions

Value is the usefulness a customer gets from a product or service. The whole customer experience (fair price, good quality, delivered on time) creates value. Companies promise value in three ways:

Strategy	Promise to the customer	Example
Operational excellence	Faster, more convenient, lower price	An e-commerce site with 1-day delivery
Product leadership	Higher-quality, better products	A flagship phone brand
Customer intimacy	Understand and respond to each customer's needs	A tailor or bank relationship manager who knows you

10. The value chain

The **value chain** is the sequence of business functions in which a product becomes more useful to customers step by step. The six links are R&D, Design, Production, Marketing, Distribution, Customer service. **Administration** (accounting, HR, IT) supports all six. Management accounting gives information to every link.

Link	Plain meaning	Example: a bicycle company
1. R&D	Generate and test new ideas	Test a lighter aluminium frame
2. Design	Detailed planning, engineering, testing	Draw the frame, test strength
3. Production	Buy, move, store, assemble	Weld frames, fit wheels
4. Marketing	Promote and sell	Ads, dealer offers
5. Distribution	Process orders, ship	Send cycles to dealers
6. Customer service	After-sales support	Free first service, warranty

11. Key success factors

Customers keep expecting better performance on five fronts: **Cost & efficiency, Quality, Time** (how fast we develop new products and how fast we respond to customers), **Innovation**, and **Sustainability** (long-term financial, social and environmental goals).

Example

A food-delivery app is judged on: price of delivery (cost), food arrives correct (quality), 30 minutes (time), new features (innovation), plastic-free packaging (sustainability).

12. The five-step decision process; planning vs control; the budget

1. Identify the problem and uncertainties.
2. Obtain information.
3. Make predictions about the future.
4. Decide by choosing among alternatives.
5. Implement the decision, evaluate performance and learn.

In simple words

Steps 1 to 4 = **Planning** (decide what to do). Step 5 = **Control** (do it, check how it went, learn and correct). The **budget** is the most important planning tool: the plan written in numbers. It helps everyone coordinate.

Example

Delivery scooter vs bike? (1) Problem: need a delivery vehicle. (2) Info: price, mileage, servicing. (3) Predict: 60 km a day for 3 years. (4) Decide: bike (cheaper to run). (5) Buy it, then every month compare the actual petrol bill with the budgeted Rs 6,000. If it is Rs 8,000, find out why. That last step is **control**.

13. Three guidelines for management accountants

1. **Cost-benefit approach:** benefits must exceed costs.
2. **Behavioural and technical considerations:** people, not just rupees, are involved.
3. **Different costs for different purposes:** compute cost differently for different decisions.

> **Example**
>
> (1) A Rs 50,000 machine saves Rs 15,000 a year for 5 years = Rs 75,000 benefit > Rs 50,000 cost: buy.
> (2) Setting an impossible sales target may look good on paper but demotivates the team (behaviour).
> (3) For pricing you may count full cost; for a one-off special order you may count only extra cost (different purpose, different cost).

14. Professional ethics

The IMA lists four standards for management accountants: **Competence, Confidentiality, Integrity, Objectivity** (also called Credibility). Ethics matter because accountants are responsible for the honesty of the numbers.

Standard	Plain meaning	Example
Competence	Keep your skills up; do the job properly	Prepare accurate reports; follow procedures
Confidentiality	Do not leak or misuse private information	Do not share salary data with a rival
Integrity	Be honest; avoid conflicts of interest; do not hide bad news	Do not hide a loss from the board
Objectivity / Credibility	Present information fairly and completely	Show both good and bad figures

Three cases from the deck:

Case	What happened	Verdict and standard
A: Leston	CFO learned a competitor was likely to go bankrupt but stayed silent in the board meeting about closing a plant.	Unethical. He withheld relevant information: **integrity / objectivity**.
B: QBX	Purchasing manager bought cheaper, lower-grade Felstar. More was used but quality was unaffected and total cost fell.	Ethical. A sound cost-benefit decision made openly for the company.
C: Central Distributing	Hillary submitted very low sales targets so she would always meet them.	Unethical. **Budgetary slack**: breach of **integrity and objectivity**.

Darius Enterprises case (QFT 3). Tae Franklin split a $250,000 remodel into $190,000 and $60,000 so it would stay under the $200,000 limit that needs board approval.

- **Effect on financial statements?** None. The total $250,000 is recorded either way. The harm is to **control**, not to the numbers.
- **Ethical?** No. He deliberately dodged an internal control, during tight cash.
- **Standards involved:** Integrity (cheating the policy), Objectivity/Credibility (the board never saw the real size), Competence (not following company procedures).

> **Example**
>
> It is like splitting one big shopping bill into two bills so you do not need your manager's signature. The amount spent is the same; the cheating is in avoiding the rule.

15. Value-added (VA) vs non-value-added (NVA) activities

> **In simple words**
>
> Ask: **"Would the customer happily pay for this step?"** Yes = value-added. No (waiting, re-doing, moving, fixing mistakes) = non-value-added. Managers try to cut NVA costs without hurting quality. The line between the two can be **subjective** (the professor's note).

Example

At a restaurant: cooking your dish is VA. Re-cooking it because the cook burnt it, or the dish waiting 20 minutes on the counter, is NVA.

(a) Falcon Enterprises (lawn mowers):

Activity	Class	Reason
1. Attaching engine to mower body	VA	Part of building the product
2. New air-conditioning for executive offices	NVA	Customer does not pay for it
3. Replacing a defective wheel	NVA	Rework: fixing a mistake
4. Designing and printing the owner's manual for a new model	VA	Customer expects and values it
5. Moving finished mowers to the warehouse	NVA	Handling adds no value
6. Attaching handle, slow due to a worker slowdown	Both	Attaching = VA; the extra time lost = NVA

(b) Company activities:

Activity	Class
Microsoft: writing code for a new spreadsheet package	VA
General Mills: painting the maintenance supervisor's office	NVA
Mayo Clinic: examining a new patient	VA
American Airlines: a 757 sitting idle 90 minutes between flights	NVA
Office Depot: moving paper cases inside the same warehouse	NVA
Rolex: attaching the band to the watch face	VA
Postal service: reprocessing wrongly sorted mail (faulty machine)	NVA
IndiaBulls: correcting staff errors in customer accounts	NVA
Marriott: upgrading bedding in very competitive markets	VA (customers value it)

(c) Two NVA costs for each service provider (deck answer):

Industry	Non-value-added costs
Airlines	Vouchers for poor service; tracing/replacing lost luggage; extra crew pay due to delays from preventable maintenance faults
Banks	Correcting bank errors; manual work during computer downtime; employee theft; loans to poor-risk borrowers that default
Hotels	Broken dishes and lost linen; replacing lost room keys; over-staffing the front desk off-peak; excess food cost

16. Class-participation answers

- **Q1:** Management accounting _____ : **(A)** focuses on estimating future revenues, costs and other measures to forecast activities and results. (B, C, D describe financial accounting.)
- **Q2:** Managers use management accounting information to _____ : **(B)** communicate, develop and implement strategies. (A, C, D are for outsiders or regulators.)

PART B: Day 2: Cost Terms, Classifications and Purposes

Big picture in one line

A **cost** is what you give up to get something. Cost accounting is just careful record-keeping of "what did it cost us to make or do this, and how does that cost change if we make more or less?"

1. What is a cost? Basic terms

Term	Plain meaning	Example
Cost	A resource given up to achieve a purpose (CAS-1: money measure of resources used for production)	Material + labour used to make an appliance
Actual cost	A cost that has already happened	You paid Rs 50,000 for steel
Budgeted cost	A predicted cost	You expect to pay Rs 55,000 next month
Cost object	Anything you want to know the cost of	A BMW X6 (product), a dealer hotline (service), an R&D project, a customer, machine set-up (activity), the safety department

Managers use cost information for two things: *making* decisions and *implementing* them.

Example

Your cost object is "one plate of biryani". Its cost = rice + chicken + cook's wage + gas + a share of the rent.

2. Cost accumulation and cost assignment

- **Cost accumulation** = collecting cost data in an organised way (the accounting system does this).
- **Cost assignment** = attaching the collected costs to a cost object, in two ways: **tracing** (direct relationship) and **allocating** (indirect relationship).

Example

Accumulation: the ledger shows rice Rs 20,000, gas Rs 5,000, rent Rs 15,000 this month. Assignment: rice goes straight to biryani (traced); rent is shared across all dishes (allocated).

3. Cost vs expense

In simple words

Cost = any sacrifice; it can sit on the balance sheet as an **asset** (stock of rice in the store). **Expense** = a cost whose benefit is used up in this period (travel, or a training course with no easy-to-measure future benefit).

Example

Rice bought Rs 10,000 and still in the store = asset (cost). The rice cooked and sold today = expense.

4. Direct and indirect costs

- **Direct cost:** you can conveniently and economically trace it to the cost object.
- **Indirect cost:** you cannot, so it is **allocated** in a reasonable, systematic way.
- Total cost of the object = direct costs (traced) + indirect costs (allocated). Managers trust direct costs more.

Direct examples	Indirect examples
Steel and tyres of a car; assembly-line wages	Electricity, rent, property tax, plant administration

Example

BMW X6: 4 tyres per car are requisitioned for that car (direct). The plant manager's salary cannot be linked to one car (indirect), so it is shared across all cars.

What decides whether a cost is direct or indirect? (1) **Materiality**: tiny costs are not worth tracing. (2) **Technology**: better systems turn more costs into direct costs. (3) **Design of operations**: a dedicated plant makes more costs direct. **Note:** a cost may be direct for one cost object and indirect for another.

Example

The salary of a supervisor who looks after only the X6 line is direct for the X6 line, but indirect for one particular car.

5. Cost behaviour: variable, fixed and mixed costs

In simple words

Variable cost = like a taxi fare: more km, more money; same rate per km. **Fixed cost** = like monthly rent: same total whatever you use, so *per-unit* cost falls as you use more.

	Total cost	Cost per unit
Variable	Rises **in proportion** to output	Stays the same
Fixed	Stays the same (in the relevant range)	Falls as output rises (changes inversely)

Example

Variable: a gumball costs Rs 0.25. Buy 100 and total = Rs 25; buy 200 and total = Rs 50; per gumball is always Rs 0.25. **Fixed (pizza):** a Rs 10 pizza cut into 4 slices = Rs 2.50 a slice; into 6 slices = Rs 1.67 a slice. The pizza's cost never changed.

Example: why per-unit fixed cost needs care

Roommate rent. Rent Rs 1,000. Two people: Rs 500 each. Three: Rs 333.33 each. Four: Rs 250 each. Trap: seeing "Rs 500 per person" and multiplying by 4 to get Rs 2,000 is wrong. Fixed costs are fixed **in total**. So "what is the fixed cost per person?" has the answer: **"it depends"** (on how many people).

Types of fixed cost: Discretionary (managers can change it in the short run: advertising, R&D) and **Committed** (long-term, cannot be cut quickly: depreciation on buildings and equipment, real-estate tax).

Relevant range: fixed and variable behaviour holds only within a normal band of activity, which is tied to **capacity**. A plane with 8 seats has a relevant range of 0 to 8 passengers; the 9th passenger needs a second plane (a new fixed cost).

Mixed (semi-variable) costs have a fixed part and a variable part: total = fixed charge + rate x usage. Examples: electricity, phone, advertising, repairs.

Example

Electricity bill = Rs 500 fixed monthly charge + Rs 8 per unit. At 100 units: 500 + 800 = Rs 1,300. At 200 units: 500 + 1,600 = Rs 2,100. To budget it you must split the bill into its fixed part (Rs 500) and variable part (Rs 8 per unit).

Why bother? Knowing cost behaviour helps us **predict, estimate, budget, plan and control** costs.

Example: fixed cost per unit (HBO slide)

The HBO bill: a monthly subscription is fixed. Watch 2 movies and each costs half the bill; watch 20 and each costs a tenth. The average cost per movie falls as you watch more.

6. Multiple classifications of a cost

A cost can be classified in two ways at once: **direct/indirect** and **variable/fixed**. That gives four combinations.

	Variable	Fixed
Direct	Steel, tyres, steering wheels of the X6	Salary of a supervisor working only on the X6 line
Indirect	Power used by shared machines (rises with output)	Plant rent, depreciation, plant administration

Exam tip / watch out

Unit-cost caveat. A unit (average) cost that includes fixed cost is true only at a stated level of output. For most decisions, think in **total costs**, not unit costs.

7. Opportunity cost and sunk cost

- **Opportunity cost** = the net benefit of the path *not* taken.
- **Sunk cost** = already spent, cannot be recovered, so it should **not** affect today's decision (can be a cost or an expense).

Example

Opportunity: you spend Saturday studying instead of working a shift that pays Rs 1,000. The Rs 1,000 is the opportunity cost of studying. **Sunk:** you bought a Rs 500 movie ticket, but the film is awful. The Rs 500 is gone either way; decide only on whether the next two hours are better spent elsewhere.

8. Decision models, relevant information and its features

- **Decision model:** a formal method of choosing between actions, using numbers and judgement.
- Good information is **Relevant, Accurate and Timely** (this is mainly the management accountant's job).
- **Relevant information** has two features: (1) it is about the **future**, and (2) it **differs between alternatives**.
- Relevant costs = expected future costs that differ; relevant revenues = expected future revenues that differ. Past (historical) costs are **never** relevant, but may help to predict.
- Items that are the same in every alternative can be ignored. The key question: **"What difference will this action make?"**
- **Quantitative factors** can be measured in numbers; **qualitative factors** (satisfaction, morale, brand image) cannot be measured easily but must still be weighed.
- Incremental costs (extra cost of an action) are basically the variable costs.

Example

Drive or take a train to a concert? Train ticket: future and different, so relevant. Annual car licence: paid either way, so irrelevant. Petrol: relevant if driving. The car's purchase price years ago: sunk, irrelevant. Qualitative factor: you can relax on the train.

9. Differential, marginal and average cost

Term	Meaning	Example
Differential cost	How much cost differs between two alternatives	Commute: Rs 300 (city) - Rs 50 (home town) = **Rs 250**
Differential revenue	How much revenue differs	Salary: Rs 2,000 - Rs 1,500 = **Rs 500**
Marginal cost	Extra cost of making **one more** unit	The 101st shirt needs Rs 300 of cloth and thread: marginal cost = Rs 300
Average cost	Total cost / number of units	Total Rs 40,000 / 100 shirts = Rs 400

Example

Job choice: home-town job Rs 1,500 with Rs 50 commute; city job Rs 2,000 with Rs 300 commute. Extra revenue Rs 500 minus extra cost Rs 250 = the city job is better by **Rs 250 a month**. Only the differences matter.

10. Class-participation answers

Question	Answer	Reason
QFT 4: Doctor says gasoline is fixed (stable price per gallon) and insurance is variable (cost per mile falls with more miles).	**Disagree on both**	Gasoline is **variable**: total rises with miles; a stable price per gallon is exactly what a variable cost looks like per unit. Insurance is **fixed**: the total is unchanged and the cost per mile falls as miles rise.
CP-3: Variable with number of cones sold?	**C and D**	Ice cream and napkins grow with cones. Lighting and manager wages do not.
CP-4: Elaine Co. total fixed cost of shipping.	**(a) $850,000**	Salaries 75% guaranteed = 0.75 x 800,000 = 600,000; plus rent 250,000 = 850,000. Packaging and postage vary with items shipped.
CP-5: Rosland Graphics lost money though prices and unit costs were unchanged.	**(b)**	Mostly fixed costs and fewer units made this year, so fixed cost per cover rose above Rs 0.27.
CP-1: Is the train ticket relevant?	**A: Yes**	Future cost that differs between alternatives.
CP-2: Is the annual car licence relevant?	**B: No**	Paid whichever way you travel.
CP-3 (sale): Car could be sold now for $5,000: sunk cost?	**B: No**	Not spent yet. It is an **opportunity cost**.

11. Types of firms and their inventories

Sector	What it does	Inventory / cost accounts	Examples
Manufacturing	Buys materials, converts them into finished goods	**Three** inventories: Raw materials (direct materials), Work-in-process (WIP), Finished goods (FG). COGS on the income statement	Microsoft, Boeing, P&G
Merchandising	Buys and resells without changing the product	**One**: merchandise inventory	Walmart, grocery, wedding-dress shops
Service	Provides intangible services	No inventory or COGS; a "cost of sales" line; maybe small supplies inventory	Audit firms, salons, FedEx

- **Raw materials**: materials in stock, ready to use. **WIP**: started but not finished. **FG**: finished and ready to sell.
- Flow of cost: **Raw materials -> WIP -> Finished goods -> COGS** (from balance sheet to income statement).

Example

Orange-juice plant. Oranges and cartons in the store = raw materials. Juice being mixed and bottled = WIP. Cartons packed in the warehouse = finished goods. Cartons sold from the shop = COGS. Until sold, the cost sits on the **balance sheet**; once sold, it moves to the **income statement**.

Gross margin statement (slide 50, with the professor's notes)

Line	Rs	Professor's note
Sales	1,000	
COGS	410	Manufacturing cost (MC); product vs period split
Gross margin	**590**	
SG&A (selling, general and administrative)	408	Non-manufacturing cost (NMC) = overheads / period costs
Operating income	**182**	

Slide detail: COGS 410 = 250 variable product + 160 fixed product. SG&A 408 = 270 variable period + 138 fixed period.

PART C: Day 3: Manufacturing Costs and the Flow of Costs

Big picture in one line

To know what a product *really* costs, add (1) the material that goes into it, (2) the wages of the people who make it, and (3) all the other factory expenses. Everything inside the factory gates is a **product cost**. Everything outside (selling, office) is a **period cost**.

1. The three manufacturing (product / inventoriable) costs

Cost	Plain meaning	Examples
Direct Material (DM)	Raw material that becomes part of the product and can be traced to it	Steel in a car; oranges, water, cartons, lids in juice
Direct Labour (DL)	Pay and benefits of people who work directly on the product	Assembly-line worker's wages
Manufacturing Overhead (MOH)	Every other factory cost that cannot be conveniently traced	See the three kinds below

Manufacturing overhead has three parts:

Part	Meaning	Examples
Indirect material	Needed for production but not an integral part of the product (or too small in value to trace)	Lubricants, cleaning supplies, glue, factory supplies like hair nets and gloves
Indirect labour	People who support production but do not work on the product	Maintenance workers, janitors, security guards, supervisors
Other indirect costs	Everything else in the factory	Depreciation on plant, factory rent or property tax, insurance, utilities, repairs, overtime premium, unavoidable idle time

Example

Orange juice carton. Direct materials: oranges, water, carton, lid. Direct labour: workers who squeeze, mix, bottle, pack. Overhead: factory depreciation (or rent), insurance, property tax, the supervisor's salary, power, repairs, hair nets and gloves.

2. Prime cost and conversion cost

In simple words

Prime cost = DM + DL (the main things you can point at). **Conversion cost = DL + MOH** (the cost of turning material into a product). DL sits in both.

Example

Clay mug. Clay Rs 20 (DM), potter's wage Rs 30 (DL), kiln gas and kiln depreciation Rs 10 (MOH). Prime cost = 20 + 30 = **Rs 50**. Conversion cost = 30 + 10 = **Rs 40**. Total product cost = **Rs 60**.

3. Non-manufacturing (period) costs

- **Selling costs:** needed to get the order and deliver the product (advertising, commission, freight outward).
- **Administrative costs:** executive, organisational and clerical costs (office salaries, office depreciation).
- Every kind of business has them ("the cost of being in business"). Whether a cost is selling or administrative can be "a matter of choice".

4. Product costs vs period costs

	Product costs	Period costs
What	DM + DL + MOH	All selling and administrative costs
First recorded in	Inventory (**balance sheet**)	Expense (**income statement**) immediately
Becomes an expense	When the product is sold (as **COGS**)	In the period it is incurred
Also called	Manufacturing / inventoriable costs	Non-manufacturing costs

Example

A factory makes 100 shirts and sells 60. Making cost Rs 100 each (Rs 10,000). Showroom rent for the month is Rs 2,000. Product cost: 60 shirts sold = Rs 6,000 goes to COGS; 40 unsold = Rs 4,000 stays in inventory (balance sheet). Period cost: the whole Rs 2,000 rent is an expense this month, even though 40 shirts are unsold.

5. The professor's "factory funnel" method: how costs flow

In simple words

Imagine a funnel. Pour in **DM, DL and MOH**. What comes out is this period's **Total Manufacturing Cost (TMC)**, also called **COP (gross)**. Then walk it through three storerooms (RM, WIP, FG). At each room, ask the same three questions: **Opening + Added = Available. Available - Closing = Used (or sent on).**

Stage	Opening + Added	= Available	- Closing	= Used / sent on
1. Direct materials	Opening + purchases + freight - returns	DM AFU (available for use)	Closing DM	**DM used** (goes into the funnel)
2. Funnel	DM used + DL + MOH	= **TMC = COP gross**		
3. Work in process	Opening WIP + TMC	WIP AFU	Closing WIP	**WIP used = COGM** ("FG made today", net COP)
4. Finished goods	Opening FG + COGM	FG AFS (available for sale)	Closing FG (balance sheet, product cost)	**COGS** (P&L)

Professor's vocabulary

Vocabulary: AFU = available for use (materials, WIP stages). **AFS** = available for sale (FG stage). **COP gross** = TMC = costs added before the WIP adjustment. **COGM / COP net** = after the WIP adjustment. Closing stock at each stage stays on the **balance sheet** as product cost; only COGS and the non-manufacturing costs reach the **P&L**.

Worked example from class: Day 1

Direct materials	Rs	Funnel (Day 1)	Rs
Opening	0	DM used	200
+ Purchase	400	DL	100
+ Freight	50	MOH	50
- Return	(25)	**COP gross = TMC**	**350**
DM AFU	**425**		
- Closing	(225)		
DM used	**200**		

Work in process	Rs	Finished goods	Rs
Opening WIP	0	Opening FG	0
+ Added (TMC)	350	+ Added (COGM)	275
WIP AFU	**350**	**FG AFS**	**275**

Work in process	Rs	Finished goods	Rs
- Completed (WIP used = COGM)	(275)	- Closing FG (balance sheet)	(50)
Closing WIP	**75**	**COGS (P&L)**	**225**

Worked example from class: Day 2 (opening = yesterday's closing)

Direct materials	Rs	Funnel (Day 2)	Rs
Opening (Day 1 closing)	225	DM used	300
+ Purchase	400	DL	200
+ Freight	100	MOH	100
- Return	(25)	**COP gross = TMC**	**600**
DM AFU	**700**		
- Closing	(400)		
DM used	**300**		

Work in process	Rs	Finished goods	Rs
Opening WIP (Day 1 closing)	75	Opening FG (Day 1 closing)	50
+ Added (TMC)	600	+ Added (COGM)	400
WIP AFU	**675**	**FG AFS**	**450**
- Completed (COGM)	(400)	- Closing FG (balance sheet)	(75)
Closing WIP	**275**	**COGS (P&L)**	**375**

In simple words

Like a water tank system: each day's **closing** level becomes the next day's **opening** level. Nothing disappears: every rupee is either used, sold, or still sitting in some stock.

Where do the "COGM" arrows go? (slide 18 to 19)

Completed goods (Rs 275 on Day 1) are called by different names depending on where you stand: **COGM** (cost of goods manufactured), **COP** (cost of production, net), **FG made today** and **COG (cost of goods) completed**. They are the same number. Add the FG opening stock to get **COG AFS**.

6. Gross margin and contribution margin income statements

	Gross margin statement (by function)	Contribution margin statement (by behaviour)
Layout	Sales - COGS = **Gross margin**; - selling and admin expenses = **Operating profit**	Sales - all **variable** costs = **Contribution margin**; - all **fixed** costs = Operating profit
Splits costs into	Product (manufacturing) vs Period (non-manufacturing)	Variable vs Fixed
Used for	External / traditional (GAAP) reporting	Internal decisions

Example

Using the slide figures. Sales 1,000. Variable: product 250 + period 270 = 520. Fixed: product 160 + period 138 = 298. **Gross margin format:** 1,000 - COGS (250 + 160 = 410) = 590; - SG&A (270 + 138 = 408) = **182**. **Contribution format:** 1,000 - 520 = 480 contribution; - 298 fixed = **182**. Same profit, two views.

Professor's order for the P&L: Sales; (-) COGS; = Gross margin (GM); (-) Sales expenses; (-) Admin expenses (both non-manufacturing); = Operating profit (OP).

Exam tip / watch out

If admin overhead is added to cost of production (old-style cost sheet), COGS and closing stock change, but total cost does not. Always say which method you are using.

7. Now You Do It: Solomon Grundy Corp. (cost flows)

No opening stocks. Tue: bought RM Rs 10,000 cash. Wed: Rs 3,600 DM moved to production. Thu: DL Rs 2,500. Fri: MOH Rs 800. Sat: finished 80% of production. Sun: sold 70% of what was finished.

Day	What happens	Balance after
Tue	Buy RM	RM = 10,000
Wed	DM to WIP	RM = 6,400; WIP = 3,600
Thu	Add DL	WIP = 6,100
Fri	Add MOH	WIP = 6,900
Sat	Finish 80%	COGM = 0.8 x 6,900 = **5,520** to FG; WIP left = **1,380**
Sun	Sell 70% of FG	COGS = 0.7 x 5,520 = **3,864**; FG left = **1,656**

- **(b) Saturday balances:** RM 6,400; WIP 1,380; FG 5,520.
- **(c) Sunday COGS:** 3,864.

8. Question for Thought 5 (Madi Corporation)

Answer

One building is used for making, selling and administration. Its depreciation should **not** be dumped as one expense. Split it among the three uses (for example by floor area). The **factory share** is manufacturing overhead (a product cost kept in inventory until sold). The **selling and administrative shares** are period costs expensed at once.

Example

Building depreciation Rs 100,000; floor area factory 80%, office 10%, showroom 10%. Factory Rs 80,000 goes into product cost; Rs 10,000 admin expense; Rs 10,000 selling expense. This is exactly how Set 1 Problem 1 splits building depreciation 8:1:1.

PART D: Day 4: Overhead Accounting, Analysis and Control

Big picture in one line

Overheads are **shared** costs (rent, power, supervisors, canteen). You cannot point at one product and say "this much rent is yours", so we **share them fairly**: first to departments, then from helper departments to the departments that make the product, and finally to the products. Like splitting a flat's electricity bill among roommates by who uses what.

1. Operating (production) vs support (service) departments

- **Operating / production department:** directly adds value to the product or service (machining, assembly).
- **Support / service department:** helps other departments (stores, maintenance, cafeteria, personnel, accounting, security, power). Its cost must be charged to the departments it serves.

Why allocate support costs? So decision-makers can see the *full* cost of each operating unit and judge its true profitability. The allocation base must truly reflect usage. A support department may serve **one** operating unit (all its cost goes there) or **two or more** (the cost must be split).

Example

The canteen costs Rs 60,000. If only Assembly uses it, all Rs 60,000 goes to Assembly. If Machining (20 staff) and Assembly (40 staff) both use it, split 1:2 = Rs 20,000 and Rs 40,000.

2. Overhead: key definitions and classification

Term	Meaning	Example
Overheads	Any cost over and above prime cost: indirect material + indirect labour + indirect expenses	Rent, power, supervisor pay
Cost allocation	Charging a **whole, identifiable** item to one cost centre or unit	Supervisor who works only in Dept A: full Rs 20,000 to A
Cost apportionment	Dividing a cost among **two or more** cost centres in proportion to benefit	Factory rent Rs 30,000 shared by floor area 2:1 = Rs 20,000 and Rs 10,000
Absorption	Charging overhead to products using a rate	Rs 5 of overhead per machine hour

Classified by: function (Production OH, Admin OH, Selling & Distribution OH); **element** (indirect material, indirect labour, indirect expenses); **behaviour** (variable, fixed, semi-variable/semi-fixed).

3. Production overhead: the five steps (we do this first)

1. **Classification** of overheads.
2. **Collection** under standing order numbers, using documents: stores requisitions (indirect material), time cards or wage analysis sheets (indirect wages), invoices / purchase vouchers, cash book (petty expenses), subsidiary records (non-cash items like depreciation and notional rent).
3. **Distribution** to production and service departments (**primary distribution**, also called departmentalisation).
4. **Redistribution** of service department costs to the production departments (**secondary distribution**).
5. **Absorption** by production units using a predetermined overhead rate.

Example

Step 1 to 5 for the rent of Rs 10,000: classify it as production overhead; collect it from the rent invoice; distribute it to departments by floor area (primary); if some is charged to the Stores department, redistribute that to the production departments (secondary); finally add it to each product through the overhead rate.

4. Primary distribution: common bases (cost drivers)

Item of expenditure	Basis of distribution	Why
Rent, depreciation, insurance (building)	Area occupied	Bigger department uses more building
Insurance and depreciation of plant & machinery; repairs	Capital value of plant	Costlier plant, more cost
Power	Horse power x machine hours	More power drawn, more cost
Heating & lighting	Floor area; light points	More space or lights, more cost
Electricity	Directly if metered; else light points	
Supervision, canteen, employer's insurance	Number of employees / wages	More staff, more cost
Fire insurance on stock	Value of stock	More stock, more risk

How to choose a base (criteria): simplicity, **availability** of data (space, equipment), and **benefit received** by the receiving department. Typical service-department bases: Personnel = employees; Receiving = units handled; Security and Custodial = square footage; Power = kWh; Cafeteria = employees; Accounting = staff hours. Rate = accumulated cost (numerator) / total of the base (denominator).

Example

Rent Rs 10,000. Areas: A 1,500, B 1,100, C 900, D 500 sq ft. Ratio 15:11:9:5 (total 40). A gets 15/40 x 10,000 = **Rs 3,750**. Rate method: 10,000 / 4,000 sq ft = Rs 2.50 per sq ft; A = 1,500 x 2.50 = Rs 3,750. Same answer. (Set 3, Problem 1 does this for every cost; see Part E.)

5. Secondary distribution: bases for service departments

- Inspection: inspection hours, number of inspectors, pieces
- Purchase: number of orders, quantity or value
- Repairs: hours worked; asset value x hours worked
- Time-keeping: number of employees, labour hours, machine hours
- Accounts: number of workers
- Stores: value or number of requisitions

In simple words

Service costs are never charged to products straight away. They are **pushed into the production departments first**, become part of each department's overhead, and then reach the product through the **predetermined overhead rate**. The deck draws this as: **First-stage allocation** = service departments to production departments. **Second-stage allocation** = production overhead (own + received) applied to products.

Example

Maintenance dept costs Rs 8,000 and does 300 hrs of work for Machining and 100 hrs for Assembly. First stage: Machining gets Rs 6,000, Assembly Rs 2,000. Second stage: if Machining's total overhead is Rs 60,000 and it runs 6,000 machine hours, the rate is Rs 10 per hour, so a product taking 2 machine hours is charged Rs 20 of overhead.

6. Single-rate vs dual-rate allocation

- **Single-rate:** does not separate fixed and variable cost; one rate per unit of one base for each cost pool.
- **Dual-rate:** splits each support department's cost into a **variable pool** and a **fixed pool** and allocates each with a different base (more accurate; can be combined with the reciprocal method).

Example

Maintenance: fixed cost Rs 60,000 (capacity) + variable Rs 2 per hour. Actual hours: X 600, Y 400; planned capacity use: X 50%, Y 50%. **Single rate:** total 62,000 / 1,000 hrs = Rs 62 per hour: X Rs 37,200, Y Rs 24,800. **Dual rate:** variable 2 x hours: X 1,200, Y 800; fixed by planned use: X 30,000, Y 30,000; total X Rs 31,200, Y Rs 30,800. Dual-rate stops X being penalised for a capacity decision shared equally.

7. Three methods of allocating service department costs

Method	How it treats services between support departments	Accuracy
Direct	Ignores them completely. Each service cost goes straight to production departments.	Simplest; widely used
Step-down	Recognises them **partly** (one-way). Start with the dept that serves the most other service depts; once allocated, nothing comes back to it.	More precise than direct; the sequence changes the answer
Reciprocal	Recognises them **fully** (two-way) using simultaneous equations (best in Excel) or repeated distribution.	Most precise; rarely used because of complexity; results close to step method

Mini example used for all three (Rs)

Service S1 costs 1,000 and S2 costs 600. S1's services are used: S2 20%, P1 40%, P2 40%. S2's services are used: S1 10%, P1 50%, P2 40%. P1 and P2 are production departments.

Method	How we do it	P1	P2	Total
Direct	Ignore S1 and S2 using each other. S1: P1:P2 = 40:40 = 500 each. S2: P1:P2 = 50:40 = 5:4, so 333.33 and 266.67	833.33	766.67	1,600
Step (S1 first)	S1 1,000: S2 200, P1 400, P2 400. S2 now = 800 split 5:4: P1 444.44, P2 355.56	844.44	755.56	1,600
Reciprocal	S1 = 1,000 + 0.1 S2; S2 = 600 + 0.2 S1. Solve: S1 = 1,081.63, S2 = 816.33. Then P1 = 0.4 S1 + 0.5 S2; P2 = 0.4 S1 + 0.4 S2	840.82	759.18	1,600

In simple words

The three answers are close but not identical. Direct ignores each service department's use of the other; step-down counts it one way; reciprocal counts it both ways. In every case the **total allocated stays the same** (1,600).

Direct method: steps

1. Take each service department's cost.
2. Ignore services between service departments.
3. Find the base ratio among **production departments only** and simplify it.
4. Multiply each share by the cost.
5. Add to each production department's primary total; check the grand total is unchanged. (Set 3, Problem 2)

Step method: steps

1. Rank service departments: first the one serving the most other service departments (or the highest cost).
2. Allocate it to **all** other departments (service and production).
3. Next department: its new total = own cost + cost received. Allocate to the **remaining** service departments and production only, never back to a closed one.
4. Continue until all service costs are in production departments. (Set 3, Problem 3)

Reciprocal method: two ways

- **Repeated distribution (cycles):** put each service total under its column, push it out by the agreed percentages, and keep repeating until the leftover is too small to matter.
- **Simultaneous equations:** write each service department's true cost as an equation (own cost + share it gets back), solve, and then distribute the true costs to production departments. (Set 3, Problem 4)

8. Administration overhead and Selling & Distribution overhead

Administration OH belongs to the business as a whole (board, legal, accounting, audit, personnel, finance, trade). It can be broken down into salaries, rent, repairs, travel, transport, telephone. It is collected under **cost account numbers**. Three ways to treat it:

Treatment	When used	Example
Transfer to P&L	It is fixed by nature and has no clear link to production or sales	Board meeting costs
Apportion to production and sales	It can be shared on an equitable basis (finding the base is the problem)	Office rent shared by floor area
Separate addition to cost of production/sales	Treat admin as its own function; base on works cost (usually best), sales value, quantity, gross profit or conversion cost	Admin OH = 10% of works cost (Set 1, Problem 3)

Selling & Distribution OH can be analysed **by nature** (freight, duty, packing, promotion, discounts), **by function** (transportation, direct selling, advertising, credit, storage), by area, product, salesman, type of customer and channel. **Control** comes from comparing with past results, budgetary control and standard costing.

Example

Selling OH of Rs 90,000 analysed **by area**: North 50,000, South 40,000. If South's sales are much lower, South's selling cost per rupee of sales is high: a control signal.

9. Allocating common costs

In simple words

A **common cost** is a cost shared by two or more users because sharing is cheaper than each paying alone (two siblings sharing one phone plan). The question is how to split the bill fairly.

Method	How it works
Stand-alone	Find what each user would pay **alone**; share the actual common cost in **proportion** to those stand-alone costs. Shared proportionally, not equally.
Incremental	Rank users. The **primary user** pays its stand-alone cost. The next user (**first incremental user**) pays the **extra** cost caused by adding it. Later users are treated the same way. The primary user usually bears the most.

Walk through 14.7: phone plan

Brother alone = 75. You alone = 85. Joint plan = 115 (saving of 45 together).

	Brother	You	Total
Stand-alone cost	75	85	160
Share (75/160, 85/160)	47%	53%	100%

	Brother	You	Total
Stand-alone allocation of 115	54.05 (47% x 115)	60.95 (53% x 115)	115
Incremental, brother primary	75	40 (115 - 75)	115
Incremental, you primary	30 (115 - 85)	85	115

In simple words

Under stand-alone, **both save** (the brother pays 54.05 instead of 75; you pay 60.95 instead of 85). Under incremental, whoever is ranked first gets no saving and the second gets all of it, which is why **ranking** matters and is often argued about.

Exam tip / watch out

The slide rounds the shares to 47% / 53%. With exact fractions, 46.875% x 115 = 53.91 and 53.125% x 115 = 61.09. Use the rounding your professor specifies.

PART E: Problem Sets: Solutions in the Professor's Method, with Excel Steps

The one idea to remember

Every cost problem tells the same story in the same order: **(1) prime cost -> (2) add factory overhead -> (3) adjust for WIP -> (4) adjust for FG -> (5) add non-manufacturing costs -> (6) compare with sales.** Learn the order once and every problem is easy.

The master skeleton (use for every problem)

Step	Name used by professor	Formula
1	DM AFU (available for use)	Opening DM + purchases + freight - returns
2	DM used	DM AFU - closing DM
3	Prime cost (PC)	DM used + DL (+ direct expenses)
4	Factory OH (MOH)	Indirect material + indirect labour + other factory costs + factory share of depreciation
5	Works cost gross = COP gross = TMC	PC + MOH
6	WIP AFU	TMC + opening WIP
7	Works cost net = COGM (WIP used)	WIP AFU - closing WIP
8	FG AFS	COGM + opening FG
9	COGS (FG used)	FG AFS - closing FG
10	Non-manufacturing (period) costs	Administration OH + selling & distribution OH
11	Total cost / cost of sales	COGS + non-manufacturing costs
12	Profit	Net sales - total cost (Gross margin = Sales - COGS; Operating profit = GM - NMC)

Exam tip / watch out

Back-solving (Set 2) runs the skeleton backwards: closing = available - used; used = available - closing; available = opening + added.

Excel basics used in all problems

1. **Labels in column A, numbers in column B.** Never type a calculated number; type a formula starting with **=**.
2. **Add/subtract:** `=B2+B3-B4`. **Total a range:** `=SUM(B2:B6)`.
3. **Percentages:** `=B10*0.4`, or format a cell as % and multiply.
4. **Lock a cell with $** when copying: `$B$2` always means B2; `C$5` locks the row; `$B7` locks the column. Press **F4** to cycle.
5. **Copy a formula:** select the cell and drag the small square at its bottom-right corner.
6. **Share by ratio:** `=C5/SUM($C$5:$F$5)*$B7` (this base / total base x cost). Check that the shares add back to the original cost.
7. **Checks:** add a cell like `=D20-SUM(D5:D8)`; it should show 0.
8. **Number format:** Ctrl+1 -> Number -> Custom -> `#,##0;(#,##0)` shows negatives in brackets. **Ctrl+B** for bold; Home -> Borders for total lines.
9. **Unknown values:** rearrange the equation on paper first (as the professor does: CC = 18 / 0.6), or use Data -> What-If Analysis -> Goal Seek.

Ready-made Excel

The companion workbook **CMA_Problem_Sets_Solutions.xlsx** has every problem below already built with live formulas: change a blue input and every answer updates.

SET 1: Cost sheets

Set 1, Problem 1: ABC Ltd cost sheet and profit/loss (year ended 31.03.2013)

In simple words

A cost sheet is a "recipe card" showing what it cost ABC to make and sell its goods. Items that are not part of running the business (interest on borrowings, bad debts, dividend income) are **financial items** and are left out.

Asked: prime cost, factory overheads, cost of production, cost of sales, and profit/loss as per cost accounts. **Key given facts:** accrued DL 8,000 and IL 1,200; depreciation on office appliances 5%, plant 10%, buildings 4%; heat-light-power 8:1:1 and building depreciation 8:1:1 (factory : office : selling); rates & taxes 2/3 factory and 1/3 office.

Professor's method

Step 1: DM used. Purchases 320,000 + freight 16,000 - returns 4,800 (a credit, so it reduces) = 331,200. Add opening 140,000 = **DM AFU 471,200**. Less closing 180,000 = **DM used 291,200**.

Step 2: DL = 160,000 paid + 8,000 accrued = **168,000**. Direct expense = 50,000.

Step 3: Prime cost = 291,200 + 168,000 + 50,000 = **509,200**.

Step 4: Factory OH (table below).

Factory overhead item	Working	Rs
Indirect labour	18,000 + 1,200 accrued	19,200
Supervision		10,000
Repairs and upkeep		14,000
Heat, light, power	8/10 x 65,000	52,000
Rates and taxes	2/3 x 6,300	4,200
Misc. factory expenses		18,700
Depreciation: plant	10% x 460,500	46,050
Depreciation: building (factory)	4% x 200,000 x 8/10	6,400
Total factory OH		**170,550**

Stage	Working	Rs
Prime cost		509,200
+ Factory OH		170,550
Works cost gross = COP gross = TMC		**679,750**
+ Opening WIP		200,000
WIP AFU		879,750
- Closing WIP		(192,000)
Works cost net = COGM	**WIP used**	**687,750**
+ Opening FG		80,000
FG AFS		767,750
- Closing FG		(115,000)
COGS	**FG used**	**652,750**

Non-manufacturing (period) cost	Working	Rs
Admin: heat, light, power	1/10 x 65,000	6,500
Rates and taxes	1/3 x 6,300	2,100
Office salaries & expenses		8,600
Depreciation: building (office)	1/10 x 8,000	800
Depreciation: office appliances	5% x 17,500	875
Total administration OH		18,875

Non-manufacturing (period) cost	Working	Rs
Selling & distribution: heat, light, power	1/10 x 65,000	6,500
Sales commission		33,600
Sales travelling		11,000
Sales promotion		22,500
Distribution salaries & expenses		18,000
Depreciation: building (selling)	1/10 x 8,000	800
Total selling & distribution OH		92,400
Total non-manufacturing cost	**18,875 + 92,400**	**111,275**

Profit and loss as per cost accounts	Rs
Sales	768,000
Less: sales returns and rebates	(14,000)
Net sales	**754,000**
Less: COGS	(652,750)
Gross margin	**101,250**
Less: non-manufacturing cost	(111,275)
Operating result: LOSS	**(10,025)**

Two layouts, same answer

The solution file's cost-sheet layout (the alternative): works cost net 687,750 + admin OH 18,875 = **cost of production 706,625** (the file prints 706,620, a typo). +80,000 - 115,000 = 671,625 (cost of production available for sale). + selling & distribution OH 92,400 = **cost of sales 764,025**. Net sales 754,000 - 764,025 = **loss 10,025**. Same loss; admin is just placed earlier.

Exam tip / watch out

Left out: interest on borrowed funds (2,000 + 2,000 accrued), bad debts 5,000, dividend income 20,000 (financial items). **Added:** accrued DL and IL. Purchase returns are a **credit** (they reduce purchases).

How to do it in Excel

Layout as in the professor's "P1s1" sheet: labels in column A, numbers in column B.

Cell	Content	Formula / value
B2:B6	Purchases, freight, returns, opening RM, closing RM	320000, 16000, 4800, 140000, 180000
B7	DM AFU	=B2+B3-B4+B5
B8	DM used	=B7-B6
B9	DL (paid + accrued)	=160000+8000
B10	Direct expense	50000
B11	PRIME COST	=B8+B9+B10
B13	Factory OH	=19200+10000+14000+0.8*65000+(2/3)*6300+18700+0.1*460500+0.04*200000*0.8
B14	Works cost gross (TMC)	=B11+B13
B15	WIP AFU	=B14+200000
B16	COGM	=B15-192000
B17	FG AFS	=B16+80000
B18	COGS	=B17-115000
B20	Admin OH	=0.1*65000+(1/3)*6300+8600+0.04*200000*0.1+0.05*17500
B21	S&D OH	=0.1*65000+33600+11000+22500+18000+0.04*200000*0.1
B22	Total non-mfg cost	=B20+B21
B24	Net sales	=768000-14000
B25	Gross margin	=B24-B18
B26	Operating profit / (loss)	=B25-B22

Steps: (1) enter inputs in cells (better than typing numbers inside formulas). (2) Build formulas top to bottom in the skeleton order. (3) Check: B11 = 509,200; B14 = 679,750; B16 = 687,750; B18 = 652,750; B26 = -10,025. (4) Format negatives in brackets.

Set 1, Problem 2: Steelsways Engineering: cost statement, estimate and price

In simple words

Use last year's accounts to learn how big overheads are *relative* to something simple (wages, works cost). Then use those percentages to cost a new machine and add profit to get the quote.

Professor's method

(a) Works cost = materials 180,000 + wages 160,000 + works overhead 40,000 = **380,000**. Total cost = 380,000 + establishment & general 19,000 = **399,000**.

Works OH % of wages = 40,000 / 160,000 = **25%**. Establishment & general % of works cost = 19,000 / 380,000 = **5%**.

(b) Materials 8,000 + wages 6,000 = prime cost 14,000. Works OH 25% of wages = 1,500. Works cost 15,500. Establishment 5% of works cost = 775. **Total cost 16,275.**

Profit 25% on cost = 4,068.75; **price = 20,343.75** (about 20,344). Check with 20% on selling price: 16,275 / 0.80 = 20,343.75. Same, because 25% on cost = 20% on price.

Exam tip / watch out

"Profit x% on cost": price = cost x (1 + x). "Profit y% on selling price": price = cost / (1 - y). 25% on cost = 20% on price.

How to do it in Excel

Cell	Item	Formula
B2:B4	Materials, wages, works OH	180000, 160000, 40000
B5	Works cost	=B2+B3+B4
B6	E&G	19000
B7	Total cost	=B5+B6
B8	Works OH % of wages	=B4/B3 (format %)
B9	E&G % of works cost	=B6/B5 (format %)
B11:B12	New machine: materials, wages	8000, 6000
B13	Works OH	=B12*B8
B14	Works cost	=B11+B12+B13
B15	E&G	=B14*B9
B16	Total cost	=B14+B15
B17	Profit 25%	=B16*0.25
B18	QUOTE PRICE	=B16+B17 (check: =B16/(1-0.2))

Set 1, Problem 3: Cost sheet with adjustments for inventories

Professor's method

Materials consumed = purchases 110,000 + opening 3,000 - closing 4,000 = 109,000. **Prime cost** = 109,000 + DL 65,000 + chargeable expenses 10,000 = **184,000**.

Factory OH 50% of DL = 32,500; **works cost gross (TMC) = 216,500**.

WIP: + opening 4,000 - closing 6,000 = -2,000; **works cost net = 214,500**.

Admin OH 10% of works cost = 21,450; **COGM = 235,950**.

FG: + opening 8,000 - closing 7,000 = +1,000; **COGS = 236,950**.

Selling OH 5% of sales 275,000 = 13,750; **cost of sales = 250,700**; **profit = 275,000 - 250,700 = 24,300**.

In simple words

Alternative (admin as period cost): COGM = 214,500; +8,000 - 7,000 gives COGS **215,500**; add selling 13,750 and admin 21,450: cost of sales **250,700**; profit **24,300**. Same profit.

How to do it in Excel

Cell	Item	Formula
B2:B4	Purchases, opening RM, closing RM	110000, 3000, 4000
B5	Materials consumed	=B2+B3-B4
B6:B7	DL, chargeable exp.	65000, 10000
B8	Prime cost	=B5+B6+B7
B9	Factory OH	=0.5*B6
B10	Works cost gross	=B8+B9
B11	Works cost net	=B10+4000-6000
B12	Admin OH	=0.1*B11
B13	COGM	=B11+B12
B14	COGS	=B13+8000-7000
B15	Selling OH	=0.05*275000
B16	Cost of sales	=B14+B15
B17	Profit	=275000-B16

Set 1, Problem 4: Cost sheet per unit (four weeks ended 30 June 2013)

Professor's method

Prime cost = 4,000 + 6,000 = 10,000 (0.50 a unit over 20,000 units).

Works OH = 1,000 machine hours x 0.50 = 500 (0.025 a unit). **Works cost** = 10,500 (0.525).

Office OH 20% of works cost = 2,100 (0.105). **Cost of production** = 12,600 (**0.63 a unit**).

Closing stock = 20,000 - 18,000 = 2,000 units x 0.63 = 1,260. **COGS** (18,000 units) = 11,340.

Selling OH = 18,000 x 0.06 = 1,080. **Cost of sales** = 12,420 (0.69). **Sales** = 18,000. **Profit = 5,580** (0.31 a unit).

Why two answers?

Alternative (office OH as period cost): closing stock = 2,000 x 0.525 = 1,050; COGS 9,450; non-manufacturing = 1,080 + 2,100 = 3,180; cost of sales **12,630**; profit **5,370**. Profit differs by 210 (= 2,000 x 0.105) because in method 1 part of the office overhead sits in closing stock instead of being expensed.

How to do it in Excel

Cell	Item	Formula
B2:B3	Units produced, sold	20000, 18000
B4:B5	RM, wages	4000, 6000
B6	Prime cost	=B4+B5
B7	Works OH	=1000*0.5
B8	Works cost	=B6+B7
B9	Office OH 20%	=0.2*B8
B10	Cost of production	=B8+B9
B11	Cost per unit	=B10/B2
B12	Closing stock value	=(B2-B3)*B11
B14	COGS	=B10-B12
B15	Selling OH	=B3*0.06
B16	Cost of sales	=B14+B15
B17	Profit	=B3*1-B16

SET 2: Back-solving inventories and absorption costing

In simple words

Some inventory figures are hidden. At every stage use the box **Opening + Added = Available; Available - Closing = Used**, and work **backwards from what you know** (the professor's "stage 1, 2, 3").

Everyday example of the box

Like finding the missing item in a bank passbook: opening 5,000 + deposits 2,000 = 7,000 available; if you know 6,500 was withdrawn, the closing balance must be 500.

Set 2, Q1: Fire at the plant: stocks on 28 Feb (Rs lakh)

Given: DM purchased 16; WIP 1 Jan 3.4; DM 1 Jan 1.6; FG 1 Jan 3.0; revenue 50; DL 18; PC 29.40; indirect factory costs = 40% of conversion cost; COG available for sale 45; gross margin 20% of revenue.

Professor's method

Stage 1 (DM): DM AFU = 1.6 + 16 = **17.6**. PC = DM used + DL, so DM used = 29.4 - 18 = **11.4**. Closing DM = 17.6 - 11.4 = **6.2**.

Find MOH: CC = DL + MOH and MOH = 0.4 CC, so DL = 0.6 CC. 18 = 0.6 CC, **CC = 30**, **MOH = 12**.

Stage 2 (funnel and WIP): COP gross = 29.4 + 12 = **41.4**. WIP AFU = 3.4 + 41.4 = **44.8**.

Stage 3 (FG, backwards from revenue): GM = 20% x 50 = 10, so COGS = **40**. FG AFS = 45, so closing FG = 45 - 40 = **5**.

Back to WIP: FG AFS = opening FG 3 + COGM, so COGM = **42**. Closing WIP = 44.8 - 42 = **2.8**.

Stock on 28 Feb	Working	Rs lakh
DM stock	17.6 - 11.4	**6.2**
WIP stock	44.8 - 42	**2.8**
FG stock	45 - 40	**5.0**

How to do it in Excel

Cell	Item	Formula
B2:B7	DM purchased, opening DM, opening WIP, opening FG, DL, PC	16, 1.6, 3.4, 3.0, 18, 29.4
B8:B10	Revenue, GM%, COG AFS	50, 0.2, 45
B12	DM AFU	=B2+B3
B13	DM used	=B7-B6
B14	Closing DM	=B12-B13
B15	CC	=B6/0.6
B16	MOH	=B15-B6
B17	COP gross	=B7+B16
B18	WIP AFU	=B17+B4
B19	COGS	=B8*(1-B9)
B20	Closing FG	=B10-B19
B21	COGM	=B10-B5
B22	Closing WIP	=B18-B21

Set 2, Q2: ABC metal shelving: DM stock, FG units, selling price, operating income

Given (Rs lakh): DM 1.4 V; DL 0.3 V; plant energy 0.05 V; indirect labour 0.1 V + 0.16 F; indirect other 0.08 V + 0.24 F; marketing/distribution/customer 1.2285 V + 0.4 F; admin 0.5 F. Variable manufacturing costs follow units **produced**; variable marketing follows units **sold**. Production 1 lakh units; 2 kg DM per unit; DM 0 kg to 2,000 kg; no WIP; revenue Rs 436,800; ending FG at the average unit manufacturing cost = Rs 20,970.

Professor's method

Step 1 (DM): used = 100,000 x 2 = 200,000 kg; purchased = 202,000 kg. The Rs 1.4 lakh is the cost of DM **used**, so Rs 0.70 a kg. **Closing DM = 2,000 x 0.70 = Rs 1,400.** (Confirmed in step 3: it gives whole FG units.)

Step 2: total manufacturing cost = 140,000 + 30,000 + 5,000 + (10,000 + 16,000) + (8,000 + 24,000) = **233,000**; per unit = **Rs 2.33**.

Step 3: FG units = 20,970 / 2.33 = **9,000**. Units sold = 100,000 - 9,000 = **91,000**.

Step 4: selling price = 436,800 / 91,000 = **Rs 4.80**.

Step 5: COGS = 233,000 - 20,970 = **212,030** (= 91,000 x 2.33).

Step 6: variable marketing = 122,850 (= 91,000 x 1.35); fixed marketing 40,000; admin 50,000.

Income statement	Rs
Revenue (91,000 x 4.80)	436,800
Less: COGS	(212,030)
Gross margin	**224,770**
Less: marketing etc. variable	(122,850)
Less: marketing etc. fixed	(40,000)
Less: administration	(50,000)
Operating income	**11,920**

Answers	
Period-ending DM inventory cost	Rs 1,400 (2,000 kg)
Period-ending FG inventory	9,000 units
Selling price per unit	Rs 4.80
Operating income	Rs 11,920

Exam tip / watch out

Fixed manufacturing costs (16,000 + 24,000) are **absorbed into inventory** in absorption costing, which is why FG carries 2.33 a unit. Marketing and admin are **period costs** and never enter inventory.

How to do it in Excel

Cell	Item	Formula
B2:B3	Units produced, kg per unit	100000, 2
B4	Closing DM kg	2000
B5	DM kg used	=B2*B3
B6	Cost per kg	=140000/B5
B7	**Closing DM cost**	=B4*B6
B9	Total mfg cost	=140000+30000+5000+10000+16000+8000+24000
B10	Per unit	=B9/B2
B11	**FG units**	=20970/B10
B12	Units sold	=B2-B11
B13	**Selling price**	=436800/B12
B14	COGS	=B9-20970
B15	Variable marketing	=122850
B16	**Operating income**	=436800-B14-B15-40000-50000

Set 2, Q3: Indirect other manufacturing cost not inventoriable; 9,000 units FG on 31 Dec

In simple words

A cost declared "not inventoriable" is a **period cost**: expensed in the year, never put in closing stock. So closing FG carries **less** per unit and profit falls.

Professor's method

Indirect other = 8,000 + 24,000 = **32,000**, expensed immediately.

Inventoriable manufacturing cost = 233,000 - 32,000 = **201,000**; per unit = **Rs 2.01**.

FG ending cost = 9,000 x 2.01 = **Rs 18,090**. **COGS** = 201,000 - 18,090 = **182,910**.

Income statement	Rs
Revenue	436,800
Less: COGS	(182,910)
Gross margin	**253,890**
Less: indirect other mfg (period cost)	(32,000)
Less: marketing variable + fixed	(162,850)
Less: administration	(50,000)
Operating income	**9,040**

Check: 11,920 - 9,040 = 2,880 = 9,000 x (2.33 - 2.01): the cost no longer carried in closing FG.

How to do it in Excel

Cell	Item	Formula
B2	Inventoriable mfg cost	=233000-32000
B3	Per unit	=B2/100000
B4	**FG ending cost**	=9000*B3
B5	COGS	=B2-B4
B6	**Operating income**	=436800-B5-32000-162850-50000

Set 2, Q4: ABC, September 2014 (Rs million)

Given: opening WIP 2,000; opening DM 900; DM purchased 3,600; DM used 3,750; variable MOH 2,500; total MOH 4,800; TMC 16,000; COGM 16,500; COGS 17,000; closing FG 1,250.

Professor's method

DM: AFU = 900 + 3,600 = 4,500; closing DM = 4,500 - 3,750 = **750**.

DL: TMC = DM used + DL + MOH, so DL = 16,000 - 3,750 - 4,800 = **7,450**.

Fixed MOH = 4,800 - 2,500 = **2,300**.

WIP: AFU = 2,000 + 16,000 = 18,000; COGM 16,500; closing WIP = **1,500**.

FG: goods available for sale = COGS + closing FG = 17,000 + 1,250 = **18,250**; opening FG = 18,250 - 16,500 = **1,750**.

Answer	Rs million
DM inventory 30 Sept	750
WIP inventory 30 Sept	1,500
FG inventory 30 Sept	1,250
Fixed manufacturing OH	2,300
Direct labour	7,450
Goods available for sale	18,250

How to do it in Excel

Cell	Item	Formula
B2:B5	Open DM, purchases, DM used, TMC	900, 3600, 3750, 16000
B6	Closing DM	=B2+B3-B4
B7	DL	=B5-B4-4800
B8	Fixed MOH	=4800-2500
B9	Closing WIP	=2000+B5-16500
B10	Goods available for sale	=17000+1250

SET 3: Overhead distribution

In simple words

Think of a hostel bill: electricity shared by room, the mess bill by number of residents, the lift by floors. Choose the **fairest base** for each cost, turn it into a ratio, simplify, split, and check that shares add back to the total.

Set 3, Problem 1: Primary distribution (A, B, C production; D service)

Costs (Rs): rent 10,000; repairs to plant 6,000; depreciation of plant 4,500; supervision 15,000; fire insurance 5,000; power 9,000; light 1,000; employer's insurance 1,500 (total 52,000).

Professor's method (ratio method)

Step 1: choose the base for each cost (area for rent; plant value for repairs and depreciation; wages for supervision; stock value for fire insurance; horse power for power; light points for light; employees for employer's insurance).

Step 2: write the ratio of the base across A : B : C : D, strike common zeros and divide by the common factor. Area 1500:1100:900:500 = 15:11:9:5 (total 40). Plant value 240,000:180,000:120,000:60,000 = 4:3:2:1 (total 10).

Step 3: share = ratio part / ratio total x cost. Rent to A = 15/40 x 10,000 = **3,750**; repairs to A = 4/10 x 6,000 = **2,400**.

Step 4: total each column; it must equal 52,000. Fire insurance is taken on **stock**, so D (no stock) gets nil.

Item	Basis	Amount	A	B	C	D
Rent	Area 15:11:9:5	10,000	3,750	2,750	2,250	1,250
Repairs to plant	Plant value 4:3:2:1	6,000	2,400	1,800	1,200	600
Depreciation of plant	Plant value 4:3:2:1	4,500	1,800	1,350	900	450
Supervision	Wages 6:4:3:2	15,000	6,000	4,000	3,000	2,000
Fire insurance	Stock 5:3:2	5,000	2,500	1,500	1,000	-
Power	HP 8:5:2	9,000	4,800	3,000	1,200	-
Light	Light points 4:3:2:1	1,000	400	300	200	100
Employer's insurance	Employees 4:3:2:3	1,500	500	375	250	375
Total		**52,000**	**22,150**	**15,075**	**10,000**	**4,775**

Rate method (the professor says *solve it both ways*): rate = cost / total base, then share = rate x base. Rent rate = 10,000 / 4,000 sq ft = Rs 2.50 per sq ft; A = 1,500 x 2.50 = 3,750. Same answer as the ratio method.

How to do it in Excel

Cell	Content	Formula
Rows 3-9	Bases for A, B, C, D (one row per base)	Enter numbers
Column F	Cost amounts	e.g. F12 = 10000
B12	Ratio method: rent to A	=B$3/SUM($B$3:$E$3)*$F12
G12	Rate method: rate per sq ft	=F12/SUM($B$3:$E$3)
B12 (rate way)	Rent to A	=$G12*B$3
B20:E20	Total per department	=SUM(B12:B19)
F20	Check	=SUM(B20:E20)-SUM(F12:F19) must be 0

Steps: (1) type bases in rows; (2) write the first formula in B12 with the $ signs exactly as shown; (3) drag it right to E12; (4) copy down and change only the base row for each cost; (5) leave blank the stock and HP cells for D so D gets nil; (6) check the total.

Set 3, Problem 2: Direct redistribution method

Primary summary: A 15,000; B 13,000; C 12,000 (40,000). Service: Stores 2,000; Time-keeping & accounts 1,500; Power 800; Canteen 500 (4,800). Total 44,800. **Bases (A:B:C):** HP 300:300:200; workers 20:15:15; stores requisitioned 2,500:1,500:1,000.

Professor's method

Rule: ignore services between service departments; split each service cost only among A, B, C.

Stores (5:3:2): 1,000 / 600 / 400.

Time-keeping & accounts (workers 4:3:3): 600 / 450 / 450.

Canteen (workers 4:3:3): 200 / 150 / 150.

Power (HP 3:3:2): 300 / 300 / 200.

Costs	Basis	Total	A	B	C
Primary distribution		40,000	15,000	13,000	12,000
Stores	Requisition value 5:3:2	2,000	1,000	600	400
Time-keeping & accounts	Workers 4:3:3	1,500	600	450	450
Canteen	Workers 4:3:3	500	200	150	150
Power	HP 3:3:2	800	300	300	200
Total		**44,800**	**17,100**	**14,500**	**13,200**

Exam tip / watch out

The solution image prints Stores to B and C as "6000" and "4000": typing slips for **600** and **400**. The printed totals are correct only with 600 and 400.

How to do it in Excel

Cell	Content	Formula
B3:D5	Bases (HP, workers, stores value)	Enter rows
E8:E11	Service costs	2000, 1500, 500, 800
B8	Stores to A	=B$5/SUM($B$5:$D$5)*$E8
B9	Time-keeping to A	=B$4/SUM($B$4:$D$4)*$E9
B12	Total for A	=SUM(B7:B11)
E12	Check = 44,800	=SUM(B12:D12)

Steps: write one formula, drag across to C and D and down, changing the base row for each service; total each production department; verify the grand total.

Set 3, Problem 3: Step method

Primary summary: X 12,000; Y 8,000. Service: Stores 2,500; Time-keeping 2,000; Maintenance 1,500. Total 26,000. **Data:** employees X 20, Y 15, Stores 10, Time-keeping 8, Maintenance 5; stores requisitions X 12, Y 10, Maintenance 3; machine hours X 1,200, Y 800.

Professor's method

Sequence: Time-keeping (serves the most), then Stores, then Maintenance.

Bases: Time-keeping by employees: Stores 10 : Maint 5 : X 20 : Y 15 (total 50; its own 8 excluded). Stores by requisitions: Maint 3 : X 12 : Y 10 (total 25). Maintenance by machine hours: X 12 : Y 8 (total 20).

Step 1: Time-keeping 2,000: Stores 400, Maint 200, X 800, Y 600. Closed.

Step 2: Stores = 2,500 + 400 = 2,900: Maint 348, X 1,392, Y 1,160. Closed.

Step 3: Maintenance = 1,500 + 200 + 348 = 2,048: X 1,228.8 (about 1,229), Y 819.2 (about 819).

Final: X = 12,000 + 800 + 1,392 + 1,229 = **15,421**; Y = 8,000 + 600 + 1,160 + 819 = **10,579**; total **26,000**.

Dept	As per PD summary	1st step (Time-keeping)	2nd step (Stores)	3rd step (Maintenance)	Final
Time-keeping	2,000	(2,000)			-
Stores	2,500	400	(2,900)		-
Maintenance	1,500	200	348	(2,048)	-
X	12,000	800	1,392	1,229	**15,421**
Y	8,000	600	1,160	819	**10,579**
Total	**26,000**				**26,000**

How to do it in Excel

Cell	Content	Formula
B4:F4	Primary totals: TK, Stores, Maint, X, Y	2000, 2500, 1500, 12000, 8000
C6:F6	Employees (Stores, Maint, X, Y)	10, 5, 20, 15
D7:F7	Requisitions (Maint, X, Y)	3, 12, 10
E8:F8	Machine hours (X, Y)	1200, 800
C10	Step 1 share	=$B$4*C6/SUM($C$6:$F$6) (drag to F10)
C11	Stores new total	=C4+C10
D12	Step 2 share	=$C$11*D7/SUM($D$7:$F$7) (drag to F12)
D13	Maintenance new total	=D4+D10+D12
E14	Step 3 share	=$D$13*E8/SUM($E$8:$F$8) (drag to F14)
E16	Final X	=E4+E10+E12+E14
F16	Final Y	=F4+F10+F12+F14
G16	Check	=E16+F16 must be 26,000

Steps: departments across the top in step order; bases in their own rows; first-step formula once with locked cost and locked total base, then drag; new total = own cost + share received; repeat for each step; check the grand total.

Set 3, Problem 4: Reciprocal method (repeated distribution and simultaneous equations)

Given: production A 8,000; B 6,500; C 7,000. Service E 2,340; F 3,000. Total 26,840. **Charge-out:** E: A 20%, B 25%, C 35%, F 20%. F: A 25%, B 25%, C 40%, E 10%. (The professor's margin ratios: E -> A 4, B 5, C 7, F 4; F -> A 5, B 5, C 8, E 2.)

In simple words

E and F **use each other**: E sends 20% of its cost to F and F sends 10% back to E. So each one's *true* cost is bigger than what is written on the page. Either solve the two equations once (exact), or keep passing the amounts back and forth until nothing is left (cycles).

Professor's method (cycles)

Method 1: Repeated distribution

1. Cycle 1: E 2,340: A +468, B +585, C +819, F +468 (F becomes 3,468).
2. Cycle 2: F 3,468: A +867, B +867, C +1,387.2, E +346.8.
3. Cycle 3: E 346.8: A +69.36, B +86.7, C +121.38, F +69.36.

4. Cycle 4: F 69.36: A +17.34, B +17.34, C +27.74, E +6.94.
5. Continue until the leftover is negligible.

Cycle	Distribute	A	B	C	Balance left
Start		8,000	6,500	7,000	E 2,340; F 3,000
1	E 2,340	8,468.00	7,085.00	7,819.00	F 3,468.00
2	F 3,468	9,335.00	7,952.00	9,206.20	E 346.80
3	E 346.8	9,404.36	8,038.70	9,327.58	F 69.36
4	F 69.36	9,421.70	8,056.04	9,355.32	E 6.94
5	E 6.94	9,423.09	8,057.77	9,357.75	F 1.39
6	F 1.39	9,423.43	8,058.12	9,358.31	E 0.14
Final		**9,423**	**8,058**	**9,358**	**total 26,839 (rounding)**

Professor's method (equations)

Method 2: Simultaneous equations (exact). Let E and F be the true costs:

- E = 2,340 + 0.10 F
- F = 3,000 + 0.20 E
- Substitute: E = 2,340 + 0.10 (3,000 + 0.20 E) = 2,640 + 0.02 E, so 0.98 E = 2,640 and **E = 2,693.88**.
- Then **F = 3,000 + 0.20 x 2,693.88 = 3,538.78**.

	A	B	C	Total
Primary distribution	8,000.00	6,500.00	7,000.00	21,500.00
From E (2,693.88) at 20% / 25% / 35%	538.78	673.47	942.86	2,155.10
From F (3,538.78) at 25% / 25% / 40%	884.69	884.69	1,415.51	3,184.90
Total after reapportionment	**9,423.47**	**8,058.16**	**9,358.37**	**26,840.00**

How to do it in Excel

Cell	Content	Formula
B2:B3	E, F primary costs	2340, 3000
B5:B6	% E gives F; % F gives E	0.20, 0.10
B8	E true cost	=(B2+B6*B3)/(1-B5*B6)
B9	F true cost	=B3+B5*B8
B11:D11	A, B, C primary	8000, 6500, 7000
B12	A total	=B11+0.20*$B$8+0.25*$B$9
C12	B total	=C11+0.25*$B$8+0.25*$B$9
D12	C total	=D11+0.35*$B$8+0.40*$B$9
E12	Check	=SUM(B12:D12) = 26,840

Matrix option: coefficients in G2:H3 = `1, -0.10 / -0.20, 1`, costs in J2:J3 = `2340 / 3000`; select two cells and enter `=MMULT(MINVERSE(G2:H3),J2:J3)`. **Cycles option:** one row per cycle, each row distributing the balance left by the row above, copied down 6 to 8 rows.

Practice problem: PlayCraft Inc. (toys)

In simple words

PlayCraft turns wood into toys. We follow the money through three storerooms: **Raw materials -> Work in process -> Finished goods**, then onto the income statement. Two traps: buying a machine or furniture buys an **asset** (only depreciation counts), and free recycled wood adds nothing.

Professor's method

Raw materials: opening 75,000 + boards 240,000 + supplies 60,000 = 375,000 available; closing 45,000; used 330,000 = DM **270,000** + indirect material 60,000.

MOH 211,000: indirect material 60,000 + supervisor 50,000 + machine depreciation 36,000 + repairs 25,000 + factory electricity 40,000.

WIP: 20,000 + 270,000 + DL 150,000 + MOH 211,000 = 651,000; closing 30,000; **COGM 621,000**. **Total manufacturing cost = 631,000**.

FG: 60,000 + 621,000 = 681,000; closing 70,000; **COGS 611,000**.

Period costs: office depreciation 8,000, admin salaries 20,000, advertising 35,000, freight outward 45,000. Machine (4,00,000) and furniture (80,000) are assets.

Raw materials inventory	Rs	Work-in-process inventory	Rs
Opening	75,000	Opening	20,000
DM purchased	240,000	+ DM used	270,000
Indirect material purchased	60,000	+ DL	150,000
		+ Manufacturing overhead	211,000
Available	**375,000**	**WIP AFU**	**651,000**
- DM used	(270,000)	- Closing WIP	(30,000)
- Indirect material used	(60,000)	**= COGM**	**621,000**
Closing RM	**45,000**		

Finished goods inventory	Rs
Opening	60,000
+ COGM	621,000
FG AFS	**681,000**
- Closing FG	(70,000)
COGS	**611,000**

(b) Required amounts	Rs
Direct material consumed	270,000
Total manufacturing costs	631,000
Cost of goods manufactured	621,000
Cost of goods sold	611,000

(c) Traditional income statement	Rs
Sales	1,200,000
Less: COGS	(611,000)
Gross profit	**589,000**
Freight outward	(45,000)
Depreciation (office furniture)	(8,000)
Administrative salaries	(20,000)
Advertising	(35,000)
Operating profit	**481,000**

Exam tip / watch out

The solution sheet prints "Direct material consumed 200,000" in part (b): a slip. Its own T-accounts give **270,000** (and 631,000 only works with 270,000).

Answer to (d)

(d) What rises with production? Variable costs rise with output: direct materials, direct labour (if piece-rate), supplies, often factory electricity. **Fixed** costs stay unchanged in the short run if capacity is enough: machine depreciation (36,000), supervisor salary (50,000). **Why it helps:** management can estimate the *extra* cost of producing more (mostly variable), see cost *per unit* fall as fixed costs spread over more units, and plan production and pricing.

Example

Fixed cost 86,000 (depreciation + supervisor) spread over 1,000 toys = Rs 86 a toy; over 2,000 toys = Rs 43 a toy. Variable cost per toy stays the same.

How to do it in Excel

Cell	Content	Formula
B2:B4	Opening RM, DM purchased, indirect material	75000, 240000, 60000
B5	Closing RM	45000
B6	DM consumed	=B2+B3-B5
B9	MOH	=B4+50000+36000+25000+40000
B10	Total manufacturing costs	=B6+150000+B9
B11	COGM	=20000+B10-30000
B12	COGS	=60000+B11-70000
B14	Gross profit	=1200000-B12
B15	Operating profit	=B14-45000-8000-20000-35000

Steps: tag every transaction first (DM / indirect material / DL / MOH / period cost / asset to ignore); enter the opening balances; build the T-accounts side by side, using SUM for each side; the closing balance is the figure that makes both sides equal.

PART F: Topic Coverage Checklist (nothing skipped)

Every slide topic of the four decks and every problem, with where it is explained.

Deck / source	Topics covered	Section
Day 1 (35 slides)	Accounting systems; language of business and users; focus of management accounting; managerial vs financial; cost vs management accounting; strategy and QFT 1; managerial accounting process; value added; value and customer value propositions; value chain; key success factors; 5-step decision process, planning, control, budget; three guidelines; ethics, cases A/B/C, QFT 2/3; value-added vs non-value-added with all exercises; CP 1 and 2	Part A, sections 1 to 16
Day 2 (50 slides)	Cost, actual, budgeted, cost object; accumulation and assignment; cost vs expense; direct/indirect and factors; variable, fixed, mixed; discretionary and committed; relevant range; per-unit fixed cost (pizza, rent, HBO); multiple classifications; unit-cost caveat; opportunity and sunk costs; decision models; relevance, qualitative vs quantitative; differential, marginal, average cost; QFT 4; CP 1 to 5; types of firms; inventories; flow of costs; gross margin statement	Part B, sections 1 to 11
Day 3 (32 slides)	DM, DL, MOH, indirect material/labour/other; selling and admin costs; product vs period costs; prime and conversion; funnel method with Day 1 and Day 2 class numbers; COGM/COP/AFU/AFS; gross margin and contribution margin statements; cost flows; Solomon Grundy; QFT 5 (Madi)	Part C, sections 1 to 8
Day 4 (56 slides)	Operating vs support departments; overhead definitions and classification; five steps of production OH; collection documents; primary distribution bases; secondary distribution; single vs dual rate; allocation bases and criteria; interdepartmental services; direct, step, reciprocal methods; repeated distribution and simultaneous equations; admin OH; selling and distribution OH; common costs (stand-alone, incremental); walk through 14.7	Part D, sections 1 to 9
Problem Set 1	Problems 1, 2, 3, 4 (both layouts)	Part E, Set 1
Problem Set 2	Q1, Q2, Q3, Q4	Part E, Set 2
Problem Set 3	Problems 1 (ratio and rate), 2 (direct), 3 (step), 4 (reciprocal, both ways)	Part E, Set 3
Extra	PlayCraft T-accounts, COGM, COGS, income statement, part (d)	Part E, end

PART G: Quick Revision Sheet

One-page formula sheet

Item	Formula
DM AFU	Opening DM + purchases + freight - returns
DM used	DM AFU - closing DM
Prime cost	DM used + DL (+ direct expenses)
Conversion cost	DL + MOH
TMC = COP gross = works cost gross	Prime cost + factory overhead
WIP AFU	TMC + opening WIP
COGM = works cost net	WIP AFU - closing WIP
FG AFS	COGM + opening FG
COGS	FG AFS - closing FG
Gross margin	Sales - COGS
Operating profit	Gross margin - (selling + admin expenses)
Total cost (cost sheet)	COGS + admin OH + selling & distribution OH
Cost per unit	Total cost / units
Fixed cost per unit	Total fixed cost / units (falls as units rise)
Differential cost	Cost of alternative 1 - cost of alternative 2

Item	Formula
Profit x% on cost	Price = cost x (1 + x)
Profit y% on selling price	Price = cost / (1 - y)
Share by ratio	(part / total of ratio) x cost
Rate method	Rate = OH / total base; share = rate x base
Reciprocal (2 service depts)	E = own_E + %F-to-E x F; F = own_F + %E-to-F x E

Common mistakes to avoid

- Putting purchase returns on the wrong side: returns **reduce** purchases.
- Forgetting **accrued** wages/expenses: add them to the paid amounts.
- Including interest, bad debts, dividend income in cost accounts: they are financial items.
- Treating a machine or furniture purchase as an expense: only **depreciation** is a cost.
- Treating selling/admin costs as product costs (unless the cost sheet adds admin into cost of production).
- Multiplying a fixed cost per unit by a new volume: fixed costs are fixed **in total**.
- Quoting a unit cost without stating the output level.
- In the step method, charging cost back to a department already closed.
- Not checking that distributed totals equal the original totals.

Glossary (A to Z)

Term	Plain meaning
Absorption	Charging overheads to products using a rate
AFU / AFS	Available for use / available for sale
Allocation	Charging a whole identifiable cost to one department or product
Apportionment	Sharing a cost between departments by benefit
Budget	The plan in numbers; main planning tool
Committed fixed cost	Long-term fixed cost you cannot cut quickly
Conversion cost	DL + MOH
Cost object	Anything we want the cost of
COGM	Cost of goods manufactured (completed in the period)
COGS	Cost of goods sold (period expense)
Differential cost	Difference in cost between two choices
Direct cost	Traceable to the cost object
Discretionary fixed cost	Fixed cost managers can change short-term
Indirect cost	Not traceable; allocated
Marginal cost	Cost of one more unit
MOH	Manufacturing overhead: all factory costs except DM and DL
NMC	Non-manufacturing (period) costs: selling and admin
NVA / VA	Non-value-added / value-added activity
Opportunity cost	Benefit of the best alternative given up
Period cost	Expensed in the period incurred
Prime cost	DM + DL
Product cost	DM + DL + MOH; held in inventory until sold
Relevant cost	Future cost that differs among alternatives
Relevant range	Activity band where cost behaviour holds
Sunk cost	Already spent; irrelevant for decisions
TMC	Total manufacturing cost = COP gross
WIP	Work in process: started, not finished